What if there was never a big bang?

What are the origins of the Universe? And have we been getting it wrong?

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The Universe is a machine for creating gods.
–Henri Bergson

Pondering the nature of existence is a fun exercise. It is at once the oldest pass time and the source of the oldest scientific questions: "How did we get here? What is time? How big is the Universe? What came before everything? What is the meaning of it all?".

The entire history of human civilisation has been littered with various answers to these questions. From religion to rationality, the spectrum of thought tackling the big questions of science and humanity have led us inextricably to one answer - there had to be a beginning. In science we call it the Big Bang. But what if that isn't it? What if there is something entirely different? Something we haven't even conceived. Well it would blow your mind, wouldn't it?

Problems with the Big Bang Theory

Scientists have since been able to trace the origins of the Universe to within less than a second after the Big Bang. However, scientists can neither explain what occurred before the Big Bang nor can they theorise with any scientific reliability the nature of the universal singularity at the time of the Big Bang.

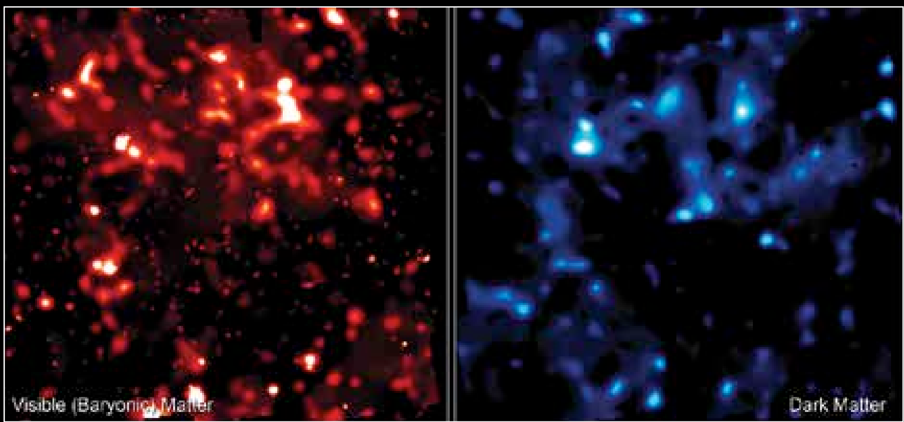
For a very long time, astronomers assumed that the Universe was composed of ordinary matter i.e. elements found on the periodic table. However, we now know that the density of ordinary matter only constitutes 4 per cent the Universe's total energy density. Of this, the stars (Hydrogen and Helium) are about 3.97 per cent, and all other heavier elements (planets, asteroids, etc., across the entire universe) a mere 0.03 per cent!

The remaining 96 per cent is made up of 23 per cent dark matter and 73 per cent dark energy. But what is dark matter and dark energy? To be honest, no one really knows. Read more about it and String Theory in this month's dmystify book.

Five dimensions?

A team of three researchers has floated a theory about the origins of the universe that challenges the traditional understanding of the Big Bang. Niayesh Afshordi, Robert Mann and Razieh Pourhasan have created a model of the universe wherein the Big Bang was in fact the implosion of a four dimensional star in a five dimensional universe. And the Universe as we see it is nothing more than the cloak or wrapping around its singularity in three dimensions. Just as a black hole's singularity is perceived to be two dimensional while in our four dimensional universe.

In their causal scenario, in a universe of four spatial dimensions (instead of our three), a four dimensional blackhole would originate in the same way due to the converging effect of gravity. This 4 dimensional black hole would also have a event horizon that cloaks it from revealing any information just as in our universe. But unlike a 3 dimensional black hole that has a 2 dimensional event horizon, a 4 dimensional black hole would have a 3 dimensional event horizon. According to these researchers our universe is actually this very 3 dimensional event horizon surrounding a 4 dimensional black hole. And just as a 3 dimensional blackhole's singularity is hidden by the 2 dimensional event horizon, so is the 4 dimensional singularity hidden by the 3 dimensional event horizon (our universe) which makes the cosmic big bang of our universe completely impossible. A brain twist, isn't it? Take a moment to read it all again.



Dark matter isn't known but it's gravitational effects tell us something is there.

Now before our collective brains explode with the idea of a five dimensional universe, consider the following - humans are used to the three spatial dimensions of length, breadth and height. Adding to it the dimension of time, we can establish that we live in a four dimensional universe. But what if there was another spatial dimension that we weren't capable of perceiving? In this model of four spatial dimensions (with time as a fifth) the researchers have proven, at least mathematically, that the new idea fits better with our observations of the cosmos than the idea of the inflationary Big Bang. The